LastSeen: Detector Comparison

CIS 5810 Final Project, Group 27
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1 Models compared

Our baseline (YOLO11s) only knows the 80 COCO classes, so it cannot find many commonly lost items such as keys, glasses or earbuds. We compared detectors along two axes: **architecture** (CNN vs. Transformer) and **vocabulary** (fixed COCO classes vs. open-vocabulary detection from a text prompt).

Table 1: Candidate detectors.

Model	Type	Vocabulary	Why we included it
YOLO11s	CNN, one-stage	COCO-80	Baseline: small and fast
Faster R-CNN	CNN, two-stage	COCO-80	Classic accurate CNN; tests a stronger CNN design
RT-DETR-L	Transformer	COCO-80	Tests the architecture with the vocabulary held fixed
YOLO-World-S	CNN + CLIP text	open	Open vocabulary at YOLO speed
YOLOE-11S / 11L	CNN + MobileCLIP text	open	Newer open-vocab YOLO; also accepts image prompts
OWLv2	ViT (Transformer)	open	Strongest open-vocab model on rare objects, but slow

We also ran YOLOE-11S at 1280 px instead of 640 px to see how much input resolution helps with small items, as a preview of SAHI tiling.

2 Evaluation

- **Data:** 9 desk/room photos and 16 query items, 8 inside COCO (phone, laptop, mouse, cup, bottle, book, plant, bowl) and 8 outside (glasses, headphones, pen, keys, earbuds case, wristwatch, camera, face mask). We labeled by hand which items appear in each photo.
- **Scoring:** a detection counts only if its box overlaps a hand-checked correct box (IoU ≥ 0.5). For each image and item, the best correct box is a hit and the best wrong box is a false positive. AP is threshold-free. Recall and precision use each model's default threshold (0.25; 0.5 for Faster R-CNN; 0.1 for OWLv2). Runtime is measured on a Colab CPU.

3 Results

Table 2: Main results on the shared 9-image benchmark (best per column in bold).

	mAP			Recall			Items	CPU
Model	all	COCO	non-COCO	COCO	non-COCO	Precision	found	ms/img
YOLO11s (baseline)	0.42	0.84	0.00	0.79	0.00	0.79	8	427
Faster R-CNN	0.45	0.91	0.00	0.79	0.00	0.73	8	11334
RT-DETR-L	0.47	**0.94**	0.00	0.91	0.00	0.72	8	1560
YOLO-World-S	0.59	0.88	0.31	0.71	0.44	0.74	12	**425**
YOLOE-11S	0.53	0.78	0.29	0.71	0.25	0.68	9	480
YOLOE-11S @1280	0.57	0.83	0.30	0.76	0.38	0.62	11	1743
YOLOE-11L	0.67	0.91	0.43	0.76	0.31	**0.82**	10	1510
OWLv2	**0.79**	0.93	**0.64**	**1.00**	**0.88**	0.43	**14**	22185

Table 3: Per-item AP. Pos. = number of images containing the item. Best per row in bold.

Item	Pos.	YOLO11s	Faster R-CNN	RT-DETR	YOLO-World	YOLOE -S	YOLOE-S @1280	YOLOE -L	OWLv2
Inside COCO									
cell phone	6	0.50	0.74	0.71	0.58	0.54	0.48	0.81	**0.86**
laptop	6	0.93	0.98	**1.00**	0.90	0.81	0.81	0.93	**1.00**
mouse	4	0.67	**0.95**	0.88	0.69	0.42	0.42	0.62	**0.95**
cup	6	**1.00**	0.92	**1.00**	0.94	0.90	**1.00**	**1.00**	**1.00**
bottle	2	**1.00**	**1.00**	**1.00**	**1.00**	**1.00**	**1.00**	**1.00**	**1.00**
book	5	0.79	0.93	0.90	0.94	0.59	**0.97**	0.93	0.66
potted plant	3	0.87	0.76	**1.00**	**1.00**	**1.00**	**1.00**	**1.00**	**1.00**
bowl	2	**1.00**	**1.00**	**1.00**	**1.00**	**1.00**	**1.00**	**1.00**	**1.00**
Outside COCO									
glasses	2	0.00	0.00	0.00	0.50	0.00	0.50	0.50	**1.00**
headphones	2	0.00	0.00	0.00	0.50	0.50	0.50	0.50	**1.00**
pen	6	0.00	0.00	0.00	0.46	0.38	0.54	0.91	**0.93**
keys	1	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
earbuds case	2	0.00	0.00	0.00	0.00	0.42	**0.83**	0.50	0.70
wristwatch	1	0.00	0.00	0.00	0.00	0.00	0.00	0.00	**0.50**
camera	1	0.00	0.00	0.00	**1.00**	**1.00**	0.00	**1.00**	**1.00**
face mask	1	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00



Figure 1: Our real photo, three detectors (boxes with confidence ≥ 0.3; blue = COCO item, red = non-COCO item). The baseline only labels COCO items (plus a false "laptop" in the corner). YOLOE-11L mislabels the sunglasses as "pen" and the charger as "camera". OWLv2 correctly finds the sunglasses and the earbuds case, but also puts a wrong "headphones" box on the keychain. No model finds the keys.

Key findings.

- **The vocabulary matters more than the architecture.** All three closed-set models (one-stage CNN, two-stage CNN, Transformer) score 0 on every non-COCO item.
- **On COCO items, stronger architectures help.** RT-DETR (0.94), OWLv2 and Faster R-CNN all beat the baseline (0.84), but Faster R-CNN is about 25× slower than the baseline.
- **Open-vocabulary models find the missing items.** OWLv2 is the most accurate overall (mAP 0.79, 14 of 16 item types found). It is the only model that found the sunglasses in our real photo (Figure 1) and the wristwatch, but it takes about 22 s per image on CPU (about 50× the baseline). YOLOE-11L (mAP 0.67) and YOLO-World (0.59, at baseline speed) offer a much better speed/accuracy trade-off.
- **Resolution helps small items.** YOLOE-11S at 1280 px raises non-COCO recall from 0.25 to 0.38, which supports adding SAHI tiling.
- **Keys and face masks are still missed by every model.** OWLv2 also has low precision (0.43): it finds more items but produces more wrong boxes.

Limitations and next steps. The benchmark is small (1–6 positive images per item), so we read trends rather than small differences. Next we will collect 30–50 of our own photos with box labels (low light, blur, clutter), add SAHI tiling, and report the same tables every iteration.